

Experiment No.: 02**Date:****Service Delivery Models (IaaS – OVHcloud)**

Assigned Date:

Submission Date:

Aim: Study and deployment of service delivery models (IaaS)

Theory: Cloud computing is a model for enabling ubiquitous, convenient, on-demand access to a shared pool of configurable computing resources. These resources can be rapidly provisioned and released with minimal management effort. It follows a utility-based model of computing where users pay only for the resources they consume, making it cost-effective and efficient.

Service Delivery Models in Cloud Computing

Cloud computing offers three primary service delivery models:

1. Infrastructure as a Service (IaaS):

Provides virtualized computing resources over the internet. Users can rent virtual machines (VMs), storage, networks, and other infrastructure components. They are responsible for managing the operating system, applications, and data.

2. Platform as a Service (PaaS):

Offers a platform that allows customers to develop, run, and manage applications without dealing with the complexity of building and maintaining the underlying infrastructure.

3. Software as a Service (SaaS):

Delivers software applications over the internet on a subscription basis. The provider manages the infrastructure and platforms that run the applications.

IaaS – Infrastructure as a Service

IaaS provides a flexible, scalable, and cost-effective solution for managing hardware resources. It offers computing infrastructure such as virtual machines, storage, load balancers, and networks. Users have full control over the operating systems and deployed applications while the service provider manages the physical infrastructure.

Key Features of IaaS

- Scalability
- Cost-efficiency (pay-per-use)

OVH → “*On Vous Héberge*” (French term). It means “**We Host You**” in English. OVHcloud is a leading cloud service provider that offers Infrastructure as a Service (IaaS) solutions to users worldwide. It enables users to create and manage virtual machines, configure storage, and set up networking components on demand. OVHcloud is known for its high-performance servers, cost-effective pricing, and strong focus on data security and privacy.

- On-demand resource provisioning
- High scalability and flexibility
- Secure infrastructure with data protection
- Cost-effective pricing model
- High availability and fault tolerance

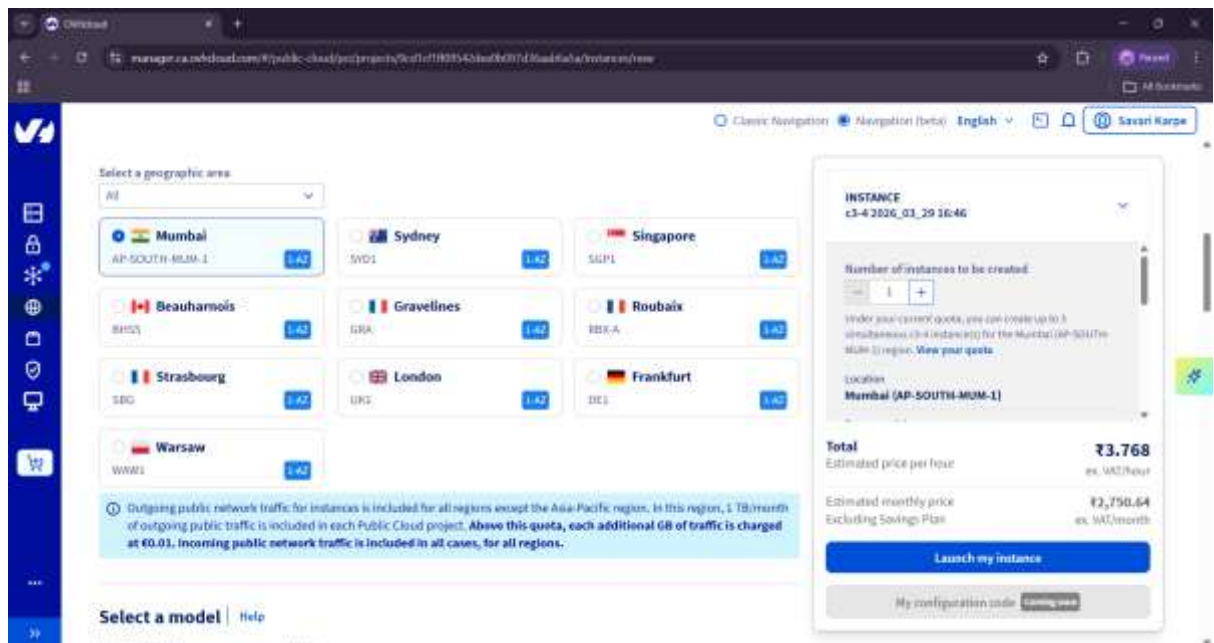
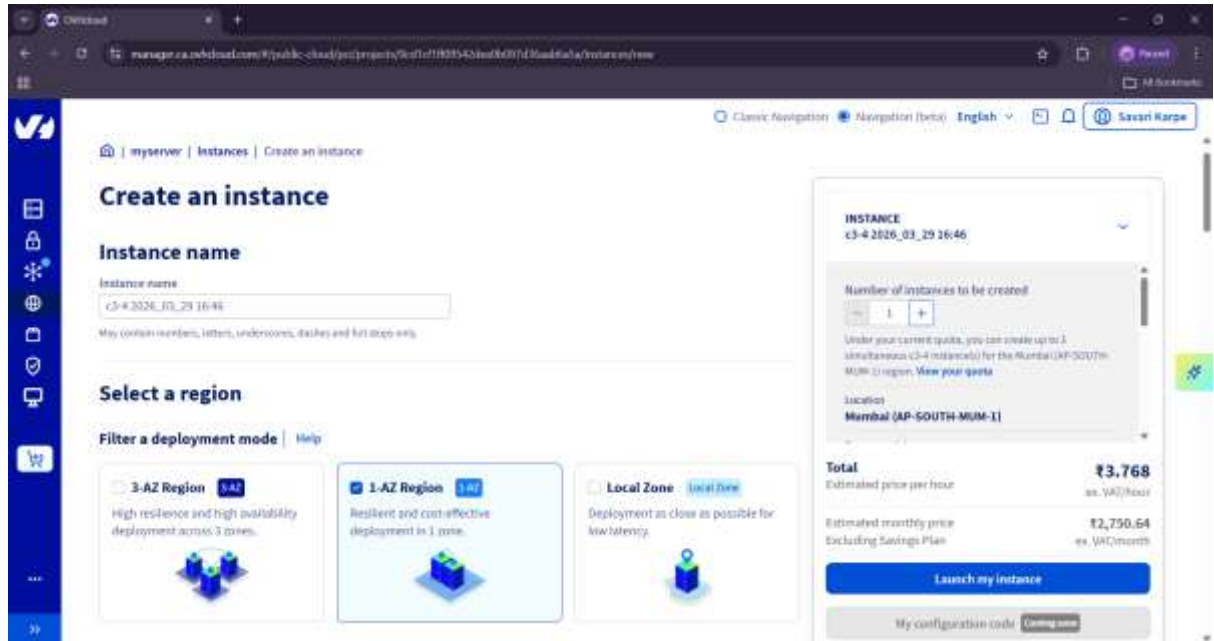
1. Login to OVHcloud Control Panel

The screenshot shows the Google Cloud Platform console interface. At the top, there's a navigation bar with 'Classic Navigation' and 'Navigation (beta)' options, along with a language dropdown set to 'English' and a user profile for 'Sauri Karpe'. The main content area is titled 'myserver' and includes a 'Roadmap & Changelog' link. A blue banner for 'Public Cloud pricing' is displayed, stating that prices for certain public cloud services are likely to change on April 1, 2026. Below this, the 'Quick access' section features six tiles: 'Instances' (Create an instance), 'Kubernetes' (Create a cluster), 'Object Storage' (Create a container), 'Block Storage' (Create a volume), 'Network' (Manage private networks (vRack)), and 'Database' (Create a database). At the bottom, an 'Other' section contains links to 'Create an AI Notebook project', 'Create a load balancer', 'Billing', and 'Quotas'.

2. Launch a Virtual Machine (Instance)

Go to the “Instances” section and click on “Create an Instance.” Choose configuration such as:

- Region (data center location)
- Select model
- Operating Image (e.g., Linux/Windows) and SSH key.



Cloud Computing Lab

Dashboard

manage.azdcloud.com/#public-cloud/projects/9c47df193543b6f007d3baad6a6a/instances/new

Classic Navigation Navigation (beta) English Savari Karpe

Select a model

Instance model: Compute Optimized Type: C5 ☒ Include unavailable

Name	Memory (GB)	vCore	Storage (GB)	Deployment	Price ex. VAT/hour	Price ex. VAT/month
c5-4	4	2	1x 30 GB NVMe	SA2	₹3.768	~ ₹2,750.64
c5-8	8	4	1x 100 GB NVMe	SA2	₹7.536	~ ₹5,501.28
c5-16	16	8	1x 200 GB NVMe	SA2	₹15.072	~ ₹11,002.56
c5-32	32	16	1x 400 GB NVMe	SA2	₹30.144	~ ₹22,005.12
c5-64	64	32	1x 400 GB NVMe	SA2	₹60.288	~ ₹44,010.24
c5-128	128	64	1x 400 GB NVMe	SA2	₹120.576	~ ₹88,020.48
c5-256	256	128	1x 400 GB NVMe	SA2	₹241.152	~ ₹176,040.96
c5-512	512	256	1x 400 GB NVMe	SA2	₹482.304	~ ₹352,081.92

INSTANCE c5-4 2026_03_29 16:46

Number of instances to be created: 1

Under your current quota, you can create up to 3 simultaneous c5-4 instances for the Mumbai (AP-SOUTH-MUM-1) region. [View your quota](#)

Location: Mumbai (AP-SOUTH-MUM-1)

Total Estimated price per hour: ₹3.768 ex. VAT/hour

Estimated monthly price Excluding Savings Plan: ₹2,750.64 ex. VAT/month

Launch my instance

My configuration code: [Getting code](#)

Dashboard

manage.azdcloud.com/#public-cloud/projects/9c47df193543b6f007d3baad6a6a/instances/new

Classic Navigation Navigation (beta) English Savari Karpe

Select an image

Distribution type: Unix distributions

To use the images, you will need to accept the supplier's user license agreement:

☒ almalinux ☒ centos ☐ cloudlinux

☐ debian ☐ fedora ☐ freebsd

☒ rockylinux ☒ ubuntu

Image version: Ubuntu 22.04

Select an SSH key

An SSH key is required to connect to your service.

Select an existing key: savari_key

+ Create a new SSH key

INSTANCE c5-4 2026_03_29 16:48

Number of instances to be created: 1

Under your current quota, you can create up to 3 simultaneous c5-4 instances for the Mumbai (AP-SOUTH-MUM-1) region. [View your quota](#)

Location: Mumbai (AP-SOUTH-MUM-1)

Total Estimated price per hour: ₹3.768 ex. VAT/hour

Estimated monthly price Excluding Savings Plan: ₹2,750.64 ex. VAT/month

Launch my instance

My configuration code: [Getting code](#)

Dashboard

manage.azdcloud.com/#public-cloud/projects/9c47df193543b6f007d3baad6a6a/instances/new

Classic Navigation Navigation (beta) English Savari Karpe

Backup settings

This feature allows you to automatically back up your instance, as often as you like.

☒ Automatic Instance Backup (Local) [Recommended](#)

Network settings

Private network

Instances are in a private network. Internet or SSH access requires a floating IP, a Gateway or an SSH Proxy.

Select an existing private network: pn-AP-SOUTH-MUM-1-Q7832026

+ Create a private network

☐ Assign a gateway - Size 5 - ₹0.243 ex. VAT / hour

Assign public connectivity

Assigning a public IP address allows access to your instance from outside the DVHcloud network.

☒ Assign a public IP

Basic Public IP (Linked to the Instance) - ₹0.50 ex. VAT / hour

A basic public IP is an address that does not persist beyond the lifetime of the instance.

☐ Assign a Floating IP (Reusable)

By selecting this option, we will assign you a 5 stored gateway.

INSTANCE c5-4 2026_03_29 16:48

Number of instances to be created: 1

Under your current quota, you can create up to 3 simultaneous c5-4 instances for the Mumbai (AP-SOUTH-MUM-1) region. [View your quota](#)

Location: Mumbai (AP-SOUTH-MUM-1)

Total Estimated price per hour: ₹3.768 ex. VAT/hour

Estimated monthly price Excluding Savings Plan: ₹2,750.64 ex. VAT/month

Launch my instance

My configuration code: [Getting code](#)

Advanced settings

Flexible instance | [Help](#)

The Flex option opts for 50GB storage, providing faster snapshots and the ability to downgrade later on.

☐ Flexible instance

Post-installation script

Enter your post-installation script

☐ Add a post-installation script

myserver | Instances

Instances

Your instance is being created, please wait a moment...

Choose an option from our catalogue to suit your requirements from the Sandbox range (shared resources), the Guaranteed Resource range (consistent performance for resource-intensive applications), the GPU range (perfect for parallel computing), and the IOPS range (for databases and big data).

You can increase the capacity of your instances at any time to suit your needs. You can also choose flex instances, which can also reduce this capacity.

Classic Navigation

Navigation (beta)

English

Save & Karp

myserver | Instances

Instances

Public Cloud pricing:
The prices of certain public cloud services are likely to change on April 1, 2026. For more details, please visit our pricing page.
[View prices](#)

+ Create an instance

Name/ID	Location	Model	Image	Public IPs	Private IPs	Volumes	Status	Actions
c3-4 2026_09_29 16:52 01a274c3-8cd4-4d97-9d1c-a623afaf94d3	Mumbai (AP-SOUTH-MUM-1)	c3-4	Ubuntu 22.04	148.113.58.8 2402:1f00:8303:750	10.1.0.46		Enrolled	

Classic Navigation

Navigation (beta)

English

Save & Karp

IaaS

Savari Sandeep Karpe

25M-CO-006

4. Connect to the Instance using SSH

```

ubuntu@07a2c65e-5ce4-49ef x + v
PowerShell 7.6.0
PS C:\Windows\System32> ssh ubuntu@148.113.58.6
The authenticity of host '148.113.58.6 (148.113.58.6)' can't be established.
ED25519 key fingerprint is SHA256:pBjHPx7svl8sluLnQF9C2KMYXP1xxydSaGVSkruJx4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '148.113.58.6' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-160-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Mar 29 11:30:54 UTC 2026

System load:  0.0      Users logged in:      0
Usage of /:   3.7% of 48.27GB  IPv4 address for ens3: 148.113.58.6
Memory usage: 6%      IPv6 address for ens3: 2402:1f00:8303::750
Swap usage:   0%       IPv4 address for ens4: 10.1.0.46
Processes:   107

```

5. Update the Server

```

ubuntu@07a2c65e-5ce4-49ef x + v
ubuntu@07a2c65e-5ce4-49ef-812a-36bf73fae8ca-0:~$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:2 http://nova.clouds.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3067 kB]
Get:5 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [437 kB]
Get:6 http://security.ubuntu.com/ubuntu jammy-security/main amd64 c-n-f Metadata [14.1 kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [5356 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-en [1025 kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 c-n-f Metadata [680 B]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1024 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/universe Translation-en [226 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 c-n-f Metadata [22.8 kB]
Get:13 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [51.9 kB]
Get:14 http://security.ubuntu.com/ubuntu jammy-security/multiverse Translation-en [10.6 kB]
Get:15 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 c-n-f Metadata [308 B]
Get:16 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:17 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3328 kB]
Get:18 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [508 kB]
Get:19 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [19.5 kB]
Get:20 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [5518 kB]
Get:21 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [1053 kB]
Get:22 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 c-n-f Metadata [676 B]
Get:23 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1261 kB]
Get:24 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [315 kB]
Get:25 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [30.4 kB]
Get:26 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [59.0 kB]
Get:27 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/multiverse Translation-en [13.5 kB]
Get:28 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 c-n-f Metadata [612 B]
Get:29 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [69.4 kB]
Get:30 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-backports/main amd64 c-n-f Metadata [412 B]
Get:31 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [30.4 kB]
Get:32 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-backports/universe Translation-en [16.9 kB]
Get:33 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 c-n-f Metadata [672 B]
Fetched 23.8 MB in 2min 29s (160 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
111 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@07a2c65e-5ce4-49ef-812a-36bf73fae8ca-0:~$

```

6. Install Web Server

Install Apache HTTP Server:

```
ubuntu@07a2c65e-5ce4-49e8-812a-36b73fae0ca-8:~$ sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils bzip2 libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.3-0 mailcap mime-support ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser bzip2-doc
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils bzip2 libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.3-0 mailcap mime-support
  ssl-cert
0 upgraded, 13 newly installed, 0 to remove and 111 not upgraded.
Need to get 2153 kB of archives.
After this operation, 8556 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64 1.7.0-8ubuntu0.22.04.2 [188 kB]
Get:2 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 amd64 1.6.1-5ubuntu4.22.04.2 [92.8 kB]
Get:3 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.2 [11.3 kB]
Get:4 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.2 [9170 B]
Get:5 http://nova.clouds.archive.ubuntu.com/ubuntu jammy/main amd64 liblua5.3-0 amd64 5.3.6-1build1 [140 kB]
Get:6 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-bin amd64 2.4.52-1ubuntu4.19 [1360 kB]
Get:7 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-data all 2.4.52-1ubuntu4.19 [165 kB]
Get:8 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-utils amd64 2.4.52-1ubuntu4.19 [89.5 kB]
Get:9 http://nova.clouds.archive.ubuntu.com/ubuntu jammy/main amd64 mailcap all 3.70+nmu1ubuntu1 [23.8 kB]
Get:10 http://nova.clouds.archive.ubuntu.com/ubuntu jammy/main amd64 mime-support all 3.66 [3696 B]
Get:11 http://nova.clouds.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2 amd64 2.4.52-1ubuntu4.19 [97.9 kB]
Get:12 http://nova.clouds.archive.ubuntu.com/ubuntu jammy/main amd64 bzip2 amd64 1.0.8-5build1 [34.8 kB]
Get:13 http://nova.clouds.archive.ubuntu.com/ubuntu jammy/main amd64 ssl-cert all 1.1.2 [17.4 kB]
```

7. Verify Apache Installation

Open browser and type: http://YOUR_SERVER_IP



8. Open Local Folder in PowerShell

Go to your folder where the index.html file is saved and in the address bar, type: powershell
Press Enter.

9. Upload HTML File to Server

Run: `scp index.html ubuntu@YOUR_SERVER_IP:/tmp`

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS D:\ME\SEM 2\CC LAB\IaaS> scp index.html ubuntu@148.113.58.6:/tmp
index.html                                100% 3651    34.3KB/s   00:00
PS D:\ME\SEM 2\CC LAB\IaaS> |

```

10. Move File to Web Directory

Go back to SSH terminal(1st powershell terminal) and
run: `sudo mv /tmp/index.html /var/www/html/`

```

ubuntu@07a2c65e-5ce4-49e0-812a-36bf73fae8ca:~$ sudo mv /tmp/index.html /var/www/html/
ubuntu@07a2c65e-5ce4-49e0-812a-36bf73fae8ca:~$ |

```

11. Restart Apache Server

Run: `sudo systemctl restart apache2`

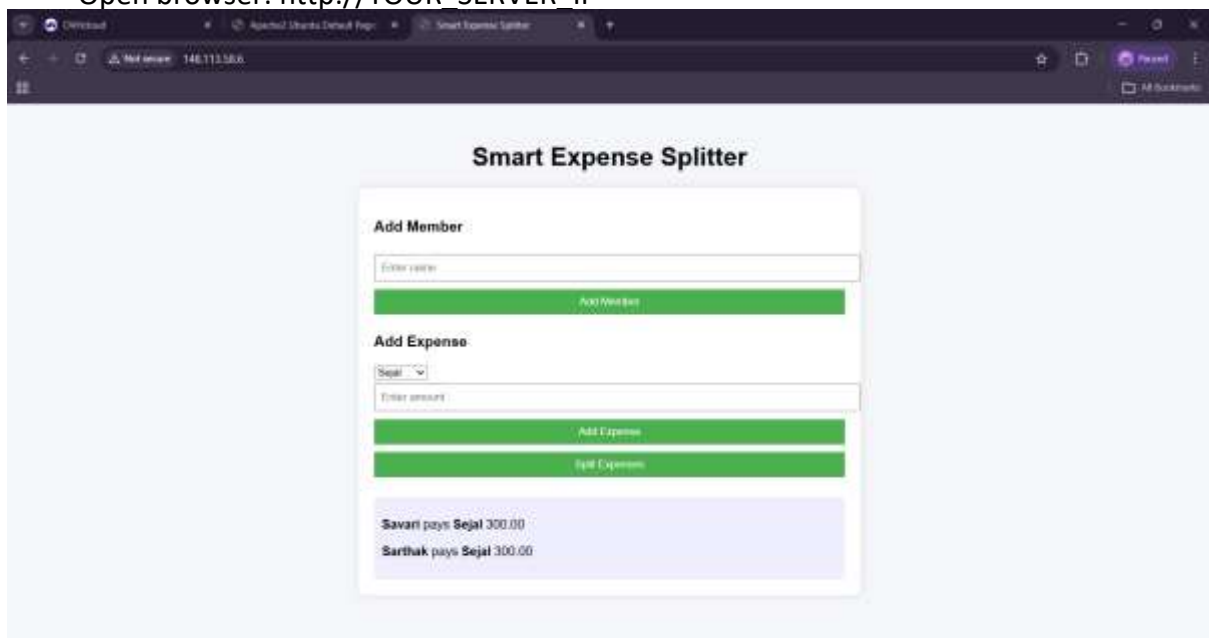
```

ubuntu@07a2c65e-5ce4-49e0-812a-36bf73fae8ca:~$ sudo systemctl restart apache2
ubuntu@07a2c65e-5ce4-49e0-812a-36bf73fae8ca:~$ |

```

12. Access the Website

Open browser: `http://YOUR_SERVER_IP`



Conclusion: Service delivery models in cloud computing provide different levels of control and flexibility to users, with IaaS being the most foundational layer. OVHcloud, as an IaaS provider, offers scalable, secure, and cost-effective infrastructure solutions. Its robust architecture, combined with advanced features and global presence, makes it a reliable choice for deploying and managing cloud-based applications efficiently.